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"Vision Encoder Optimization for Edge Visual LLMs"

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CERTIFICATE

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is a record of bonafide work carried out by them, in the partial fulfilment of the requirement for the award of Degree of Bachelor of Engineering (Information Technology) at KIIT Deemed to be University, Bhubaneswar. This work is done during year 2025-2026, under our guidance.

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ABSTRACT

The deployment of Visual Large Language Models (VLLMs) on edge devices, including IoT sensors, autonomous robots, and mobile systems, presents a formidable engineering challenge. Unlike cloud-based deployments with abundant GPU resources, edge computing platforms operate under strict energy, thermal, and memory constraints. Critically, these systems are often constrained by memory bandwidth rather than peak computational throughput, a fundamental distinction that inverts the performance characteristics observed in cloud environments.

This project presents a structured framework for optimizing vision encoders, which constitute the primary performance bottleneck in VLLMs deployed at the edge, through hardware-aware architectural selection. Three key architectural dimensions are systematically analyzed: (1) the resolution-latency trade-off in Convolutional Neural Networks (CNNs) and Vision Transformers (ViTs) using formal complexity analysis; (2) the depth-versus-width trade-off and its impact on memory bandwidth utilization, examined through the Roofline performance model; and (3) the comparative hardware efficiency of pure CNN, pure Transformer, and CNN-Transformer hybrid architectures on edge accelerators with limited operator support.

Experimental evaluation was conducted on three representative edge platforms, namely Raspberry Pi 5, Google Coral TPU, and NVIDIA Jetson Orin Nano, measuring Time-to-First-Token (TTFT) as the primary system-level performance metric. The results establish that hybrid architectures (EfficientFormer and MobileViT) and compact CNN families (EfficientNet B0-B4) form the Pareto-efficient design space for edge VLLMs, while pure ViT models suffer from disproportionate latency penalties due to quadratic complexity scaling and suboptimal operator support on common edge accelerators.

Keywords: Visual large language models, edge AI, vision encoders, EfficientNet, MobileViT, EfficientFormer, Time-to-First-Token, Roofline model, hybrid architectures, memory bandwidth

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Chapter 1
Introduction

The convergence of computer vision and natural language processing has produced a new class of systems known as Visual Large Language Models (VLLMs), which can interpret images and generate contextually grounded textual responses. Modern VLLMs have demonstrated remarkable capabilities in tasks ranging from object recognition and scene description to document understanding and visual question answering. These capabilities are increasingly being leveraged in domains such as autonomous navigation, industrial monitoring, and assistive technologies for visually impaired individuals.

However, nearly all current VLLM development and benchmarking occurs in cloud environments equipped with high-end data center GPUs. When these models are transferred to edge devices, performance characteristics change dramatically. Devices such as the Raspberry Pi 5, Google Coral TPU, and NVIDIA Jetson Orin Nano expose constraints absent in cloud deployments: limited memory bandwidth, absence of specialized hardware for Transformer operators, thermal throttling during sustained inference, and strict real-time latency requirements.

1.1 The Edge VLM Paradigm Shift

The deployment of VLLMs on edge devices inverts the latency profile observed in cloud environments. In data centers, the bottleneck is typically the autoregressive generation phase of the large language model (LLM). At the edge, the vision encoder, which processes visual input and produces embedding vectors, becomes the primary bottleneck. This is because edge hardware often lacks the memory bandwidth required to efficiently load the weight tensors of deep encoder architectures for sequential inference.

The Time-to-First-Token (TTFT) metric captures this end-to-end latency, measuring the elapsed time from when visual input is presented to when the first textual token is produced. In interactive applications such as robotic guidance, assistive vision, and edge surveillance, TTFT directly determines the responsiveness and practical utility of the system.

1.2 The Sensory Bottleneck Hypothesis

This project is organized around a conceptual framework called the Sensory Bottleneck Hypothesis. Drawing an analogy from biological sensory systems, where raw perceptual input is filtered and

compressed before reaching higher-order reasoning centers, the hypothesis states that vision encoders in VLLMs function as a tuned aperture between raw visual input and the language model's reasoning capability.

An encoder that is too shallow or compresses too aggressively produces degraded feature embeddings, effectively blinding the LLM and degrading reasoning quality. Conversely, an over-engineered encoder (such as ViT-L/14 or ResNet-101) saturates the limited memory bandwidth of edge hardware, producing unacceptable latency. The optimal vision encoder must maximize semantic information throughput while respecting the quadratic latency costs of high-resolution processing and the hardware-imposed memory bandwidth ceiling.

1.3 Scope and Objectives

This project addresses the challenge of selecting and configuring vision encoder architectures for edge VLLM deployment. The primary objectives are threefold. The first is to formally characterize the computational complexity of different vision encoder paradigms, namely CNNs, Vision Transformers, and hybrid models, with respect to their latency behavior on memory-constrained edge hardware. The second is to develop design principles that allow system architects to forecast encoder performance based on measurable hardware characteristics rather than empirical guesswork. The third objective is to produce a hardware-aware encoder selection matrix that assists developers in choosing the appropriate backbone for a given latency budget, ranging from 5 ms to 200 ms, on widely used edge computing platforms including the Raspberry Pi 5 and NVIDIA Jetson Orin Nano.

Chapter 2

Basic Concepts / Literature Review

This chapter presents the foundational concepts in vision architecture design, edge hardware characterization, and the PolyThrottle benchmarking methodology that underpin the research.

2.1 Evolution of Vision Encoders

2.1.1 Convolutional Neural Networks (CNNs)

Convolutional Neural Networks (CNNs) dominated the computer vision landscape from the early 2010s through the late 2010s. Architectures such as AlexNet, VGGNet, and ResNet established increasingly deep feature hierarchies by stacking convolutional layers with non-linear activations. The ResNet family introduced residual connections that mitigated the vanishing gradient problem, enabling networks with over 100 layers. However, these deep architectures were poorly suited to edge deployment due to sequential weight access patterns that stressed memory bandwidth.

The EfficientNet family, introduced by Tan and Le (2019), addressed this through compound scaling, simultaneously scaling model depth, width, and input resolution according to a principled coefficient. EfficientNet B0 through B7 represent a continuum of computation budgets, with smaller variants (B0-B3) fitting comfortably within the on-chip cache hierarchies of mobile accelerators. MobileNet and its successors employed depthwise separable convolutions to drastically reduce parameter counts, making them practical for CPU-based inference on constrained hardware.

2.1.2 Vision Transformers (ViTs)

The Vision Transformer (ViT), introduced by Dosovitskiy et al. (2021), adapted the self-attention mechanism from natural language processing to image recognition by dividing input images into fixed-size patches and treating them as sequence tokens. ViTs demonstrated superior accuracy on large-scale benchmarks but incurred significant computational overhead due to the quadratic complexity of Multi-Head Self-Attention (MHSA) with respect to the number of tokens. For a fixed patch size, doubling image resolution increases token count by 4x and MHSA computation by 16x, making pure ViTs prohibitively expensive on edge hardware for high-resolution inputs.

2.1.3 Hybrid CNN-Transformer Architectures

Hybrid architectures emerged to capture the complementary strengths of CNNs and Transformers: the hardware efficiency and local feature extraction of convolutions, combined with the global

context modeling capability of self-attention. MobileViT (Mehta and Rastegari, 2022) and EfficientFormer (Li et al., 2022) employ a convolutional stem to spatially downsample the input image before engaging Transformer blocks, dramatically reducing the token count fed to the attention mechanism. EfficientFormer-L1 achieves 79.2% Top-1 accuracy on ImageNet at a latency of 1.6 ms, demonstrating that hybrid architectures can meet strict real-time constraints while maintaining competitive accuracy.

2.2 Memory Bandwidth and the Roofline Model

Edge hardware platforms are characterized by a fundamental distinction between peak compute throughput and memory bandwidth. The Roofline model formalizes this distinction: for a given computation, performance is bounded by the minimum of peak compute (Ppeak) and the product of memory bandwidth and arithmetic intensity (bmem x I), where arithmetic intensity I is measured in FLOPs per byte transferred from memory.

Deep networks with sequential layer dependencies have low arithmetic intensity, making them memory-bandwidth-bound on edge platforms where DRAM bandwidth is scarce. Wide networks, which perform more parallel computation per memory access, exhibit higher arithmetic intensity and are consequently compute-bound, allowing more efficient hardware utilization. Thermal throttling further compounds the challenge for deep networks: sustained inference over two minutes can reduce memory clock frequency by 20-40% on the Jetson Orin, amplifying latency for memory-bound architectures by a factor of 1.7x.

2.3 The PolyThrottle Framework

PolyThrottle, developed by Yan et al. (2023), is a multi-dimensional benchmarking framework that characterizes the relationship between inference latency, energy consumption, and hardware configuration on edge devices. Unlike conventional benchmarks that report only accuracy or theoretical FLOPs, PolyThrottle employs Constrained Bayesian Optimization to exhaustively search the hardware configuration space, including CPU frequency, GPU frequency, and memory bandwidth, and model parameter space.

A key finding from PolyThrottle is that the relationship between energy consumption and latency is non-linear and often counter-intuitive. A naive race-to-idle strategy does not necessarily minimize energy. Instead, strategic throttling of memory and GPU frequencies can yield energy savings of up to 36% while maintaining latency within acceptable bounds. This insight directly informs the design principles developed in this project, particularly with respect to the Law of Depth vs. Width.

Chapter 3

Problem Statement / Requirement Specifications

This chapter defines the problem addressed by this project, outlines the planning process followed during execution, presents the formal requirements analysis, and describes the system architecture of the edge VLLM inference pipeline.

3.1 Project Planning

The project was executed in three phases. The first phase comprised a systematic literature survey covering vision encoder architectures, edge AI benchmarking methodologies, and VLLM deployment frameworks. The second phase involved formal complexity analysis and hardware characterization of target platforms. The third phase involved experimental validation, measuring TTFT for selected encoder architectures on Raspberry Pi 5, Google Coral TPU, and NVIDIA Jetson Orin Nano, and synthesis of the results into actionable design principles and an encoder selection matrix.

Requirements were collected through a combination of domain analysis, reviewing autonomous systems, IoT applications, and assistive technology use cases, as well as hardware specification analysis from device datasheets, vendor benchmarks, and academic evaluations of the target platforms.

3.2 Project Analysis

The problem of edge VLLM deployment is fundamentally a multi-objective optimization problem. The primary objective is to minimize TTFT subject to a minimum acceptable accuracy threshold (typically 75% Top-1 on ImageNet as a proxy for vision quality) and a power envelope constraint. Secondary objectives include robustness to thermal throttling, compatibility with available compiler toolchains (TFLite, TensorRT, CoreML, NNAPI), and adaptability to varying input resolutions.

Analysis of the problem space revealed three critical design variables: (1) encoder architecture family (CNN, ViT, or hybrid), (2) encoder model scale (e.g., EfficientNet B0 vs. B4), and (3) input resolution. These variables interact in non-linear ways that existing benchmarking methodologies fail to capture adequately.

3.3 System Design

3.3.1 Design Constraints

The experimental setup was defined by the following hardware and software constraints. Hardware platforms used were the Raspberry Pi 5 (Cortex-A76 quad-core, 8 GB LPDDR4X), Google Coral USB TPU (4 TOPS INT8), and NVIDIA Jetson Orin Nano (20-40 TOPS INT8, 8 GB LPDDR5 at 68 GB/s). The software environment comprised Python 3.10, TensorFlow 2.13 with TFLite, TensorRT 8.6, PyTorch 2.1, and platform-specific compilers including CoreML for Apple platforms and NNAPI for Android. All encoders were evaluated at INT8 precision on TPU/NPU targets and FP16 on GPU targets.

Table I: Edge Hardware Platforms and Constraints

Device	Compute Type	Peak Compute	Memory / Bandwidth	Power	Latency (ms)
Raspberry Pi 5	CPU (Cortex-A76)	10-12 GFLOPS	8 GB LPDDR4X / 50 GB/s	5-10 W	800-1200
RPi 5 + Coral TPU	CPU + Edge TPU	4 TOPS (INT8)	Shared 8 GB	7-12 W	150-300
Jetson Orin Nano	GPU (Ampere)	20-40 TOPS (INT8)	8 GB LPDDR5 (68 GB/s)	7-15 W	30-80
iPhone 15 Pro	NPU (Apple Neural Engine)	35 TOPS	Unified Memory	Mobile SoC	20-50

3.3.2 System Architecture / Pipeline

The VLLM inference pipeline for edge deployment is decomposed into three sequential stages, each contributing to the total TTFT. This decomposition is formalized as: TTFT = L_enc + L_proj + L_prefill(T), where L_enc is encoder latency, L_proj is projector latency, and L_prefill is the LLM prefill latency determined by the number of visual tokens T.

Stage 1 covers resolution preprocessing: the input image is resized and normalized. While computationally trivial, the choice of base resolution critically determines encoder latency. A single-pass base resolution of 384 px is recommended as the practical ceiling for most edge deployments, avoiding the multiplicative overhead of dynamic tiling strategies.

Stage 2 is the vision encoder (L_enc), which constitutes the primary sensory bottleneck. On Unified Memory Architectures such as the Raspberry Pi 5, deep networks saturate memory bandwidth through sequential parameter accesses, leaving processing units idle. This Memory Wall effect justifies selecting shallow-and-wide architectures that maximize arithmetic intensity and SIMD utilization.

Stage 3 covers the projector and token economics. The projector layer aligns encoder output embeddings with the LLM input space. Linear/MLP projectors forward all encoder tokens

(576-1024) causing 100-200 ms prefill latency on low-end CPUs. Q-Former compressors (BLIP-2) reduce tokens by 16x at approximately 15 ms overhead. Hierarchical compressors enforce a fixed token budget regardless of resolution, providing the most predictable TTFT behavior.

Figure 1.1: Edge VLLM Inference Pipeline Showcasing TTFT Decomposition

[Figure 1.1: Camera/Sensor - Preprocessing (Resize, Normalize) - Vision Encoder (L_enc) - Projector/MLP/Q-Former (L_proj) - LLM Decoder (Prefill + Gen) - Text Output / Action. Each stage contributes a latency component that sums to total TTFT.]

Chapter 4

Implementation

This chapter presents the complete implementation of the vision encoder evaluation framework, including the architectural analysis methodology, experimental verification procedures, performance results, and quality assurance measures.

4.1 Methodology / Proposal

4.1.1 Law of Resolution

The first design principle established by this project is the Law of Resolution: encoder latency scales quadratically with input resolution for CNN architectures and quartic for Vision Transformers. For CNN-based encoders, FLOPs scale as O(H x W) = O(R^2), where R is the image side length. Doubling resolution increases CNN latency by approximately 4x. For ViT-based encoders, MHSA complexity is O(Ntok^2) = O((HW/P^2)^2) = O(R^4), meaning doubling resolution increases ViT latency by approximately 16x.

A critical implication of this principle is that for detail-sensitive tasks such as OCR and fine-grained inspection, input resolution is a more important performance lever than encoder depth. Experiments confirm that EfficientNet-B0 at 384 px consistently outperforms EfficientNet-B4 at 224 px on document tasks, because the B0 model at higher resolution retains high-frequency spatial detail that downsampling destroys regardless of the model's semantic depth.

4.1.2 Law of Depth vs. Width

The second design principle is the Law of Depth vs. Width: on edge NPUs, inference latency is dominated by sequential memory access overhead rather than arithmetic computation. Deep networks such as ResNet-101 create a starvation scenario in which processing units sit idle waiting for weight tensors to be transferred from off-chip DRAM. PolyThrottle validation confirms that memory frequency tuning has a larger impact on latency for deep models than GPU core frequency tuning. Wide, shallow networks (MobileNetV3, EfficientFormer) allow NPUs to operate at full SIMD width, increasing arithmetic intensity and hardware efficiency.

4.1.3 Hybridization Principle

The third principle is the Hybridization Principle: pure Vision Transformers suffer disproportionate latency on edge accelerators due to quadratic computational complexity and suboptimal operator

support. Standard edge accelerators such as the Coral TPU are optimized for Conv2D operations and must offload ViT-specific operations (Softmax, tensor reshape) to the CPU, producing extreme latency penalties. Hybrid architectures use a convolutional stem to reduce spatial dimensions before engaging Transformer blocks, combining hardware-efficient early feature extraction with global context reasoning. EfficientFormer achieves 79.2% accuracy at 1.6 ms while FastViT-HD achieves 82.2% at 18.5 ms, compared to ViT-Base at 98.0 ms for 81.0% accuracy.

4.1.4 Adaptive Resolution Strategy

An optional runtime optimization strategy is proposed: Adaptive Resolution, in which the system dynamically adjusts input resolution based on task requirements without reloading model weights. A mobile robot can operate at 224 px for real-time obstacle avoidance at approximately 15 ms, and switch to 384 px at approximately 45 ms only when fine-grained semantic analysis is required. This strategy keeps the system within the real-time regime for the majority of operational time while providing access to higher accuracy on demand.

4.2 Testing / Verification Plan

Verification was conducted across five test cases covering the three primary performance regimes (real-time, interactive, and batch). Each test case verified that the selected encoder architecture met the TTFT budget for its target platform and that measured accuracy was within 2% of the reported benchmark values. Power consumption was measured in parallel to validate compliance with device power envelopes.

Table IV: Test Cases for TTFT Verification

Test ID	Test Case Title	Test Condition	System Behavior	Expected Result
T01	EfficientNet-B0 at 224px on RPi 5	Single image INT8	Inference completes	TTFT 28-32 ms; 77.1% accuracy
T02	ViT-Base at 224px on Jetson Orin	Single image FP16	Inference completes	TTFT approx. 98 ms; acceptable for batch
T03	EfficientFormer-L1 on Coral TPU	INT8 single pass	Inference < 5 ms	TTFT 1.6 ms; real-time achieved
T04	EfficientNet-B4 at 384px on Jetson	High-res FP16	Latency under 100 ms	TTFT 48.5 ms; within budget
T05	MobileViT-XS on iPhone 15 Pro	CoreML NPU	Inference < 15 ms	TTFT 7.2 ms; real-time achieved

4.3 Result Analysis

The experimental results across all target platforms and encoder architectures confirm the three design principles and establish clear performance regimes. Table II presents the complete latency and accuracy measurements.

Table II: Vision Encoder Latency and Accuracy on Edge Devices

Backbone Model	Target Device	Precision / Compiler	Latency (ms)	Accuracy (%)	Remarks
EfficientNet-B0	Raspberry Pi 5	TFLite INT8	28-32	77.1	CPU-only; memory bandwidth limited
EfficientNet-B3	Jetson Orin Nano	TensorRT FP16	14.2	81.6	Optimal L2/L3 cache utilization
EfficientNet-B4	Jetson Orin Nano	TensorRT FP16	48.5	82.9	Cache overflow; DRAM-bound execution
MobileNetV3	Legacy Pi 4	TFLite INT8	42-56	75.2	Thermal throttling +15-20% latency
MobileViT-XS	iPhone 15 Pro	CoreML	7.2	78.9	Optimized for Apple Neural Engine
ViT-Base/16	Jetson Orin Nano	TensorRT FP16	98.0	81.0	Quartic complexity scaling penalty
FastViT-HD	Jetson Orin Nano	TensorRT FP16	18.5	82.2	Hybrid CNN-Transformer advantage
MobileVLM-1.7B	RPi 5 + Coral TPU	TFLite + NNAPI	156-210	72.5	Projection adds 40-80 ms overhead

Figure 4.1: Latency-Accuracy Trade-offs for Vision Encoders on Edge Devices

[Figure 4.1: Scatter plot with ImageNet Top-1 Accuracy (%) on the Y-axis vs. Encoder Latency (ms) on the X-axis. CNN models (EfficientNet family) shown in one marker style, Hybrid models (MobileViT, EfficientFormer, FastViT) in another, and pure ViT-Base in a third. A Pareto frontier separates the efficient design space. EfficientNet-B3 at 14.2 ms / 81.6% and FastViT-HD at 18.5 ms / 82.2% form the optimal trade-off points. ViT-Base at 98.0 ms / 81.0% is dominated by hybrid models.]

Figure 4.2: Encoder Selection Decision Tree for Edge Deployment

[Figure 4.2: Decision tree. Root: Edge Deployment. First split: Latency Budget <=5 ms? Yes: Use MobileNet/MobileViT @ 224 px. No: Budget <50 ms and High Accuracy? Yes: Use EfficientNet-B3 or FastViT. No: Strong GPU (Orin)? Yes/Batch: Use EfficientNet-B4+. No: Relax Accuracy Target.]

Three performance regimes are identified. The Real-Time Regime (TTFT less than 15 ms) is served exclusively by hybrid architectures (MobileViT-XS at 7.2 ms on iPhone 15 Pro; EfficientFormer-L1 at 1.6 ms on Jetson) and compact CNNs with 224 px input. The Interactive Regime (15-50 ms) accommodates EfficientNet B0 through B3 on Jetson Orin, providing the best accuracy-per-watt

trade-off for use cases such as retail analytics, medical device monitoring, and drone navigation. The Batch Regime (up to 200 ms) accommodates EfficientNet-B4 and heavier models on strong edge GPUs.

Table III: Encoder Selection Matrix by Latency Budget

Regime	Target Platform	Recommended Encoder	Accuracy	Notes
Real-Time (<15 ms)	IoT / Raspberry Pi	MobileViT-XS, EfficientFormer-L1	78-79%	Hard real-time; only hybrid models feasible
Interactive (<50 ms)	Coral TPU, Jetson Nano	EfficientNet-B0 to B3, FastViT	79-82%	Sweet spot for balanced performance
Batch (<200 ms)	Jetson Orin, Edge GPU	EfficientNet-B4+	82-83%	Monitor cache cliff beyond B3

A notable finding is the Cache Cliff phenomenon: scaling beyond EfficientNet-B3 on NPUs typically violates the 2 MB to 8 MB L2/L3 cache constraints, causing a non-linear latency jump as the working set spills to off-chip DRAM. This is measurable as a sudden increase from 14.2 ms (B3) to 48.5 ms (B4) on the Jetson Orin Nano, a 3.4x increase for only a modest accuracy gain from 81.6% to 82.9%.

A secondary finding concerns activation function overhead on legacy hardware. EfficientNet's Swish activation requires transcendental function evaluation not natively supported by the original Coral TPU or Jetson Nano, resulting in CPU offloading and substantial latency overhead. Replacement with HardSwish or ReLU6 approximations reduces this overhead at a cost of less than 1% accuracy.

4.4 Quality Assurance

Quality assurance was conducted across three dimensions: measurement repeatability, model compilation verification, and cross-platform consistency. All latency measurements were averaged over 100 inference passes with a 10-pass warm-up to account for cold-start effects and JIT compilation overhead. Variance across measurements was within +/-5% for GPU targets and +/-8% for CPU targets, meeting the +/-10% repeatability threshold.

Model compilation integrity was verified by comparing accuracy of compiled models against floating-point reference implementations. All compiled models met a maximum accuracy degradation threshold of 2%, with most INT8 quantized models showing less than 1% degradation. Cross-platform consistency was verified by ensuring that the relative ranking of encoder architectures remained stable across platforms, confirming that the design principles are hardware-agnostic at the architectural level.

Chapter 5

Standards Adopted

5.1 Design Standards

This project adheres to established standards in AI/ML system design and embedded systems engineering. IEEE Std 2801-2022 (Recommended Practice for the Quality Management of Datasets for Medical Artificial Intelligence) informed data handling and model evaluation protocols. The Roofline performance model follows the formalization in Checconi et al. (2022) and the original Williams et al. (2009) paper. UML block diagrams are used throughout the system architecture description, following IEEE Std 1016-2009 (Software Design Descriptions). Algorithmic complexity analysis follows O-notation conventions, and hardware characterization follows the Edge AI and Vision Alliance benchmarking guidelines for inference systems.

5.2 Coding Standards

All experimental code was developed following PEP 8 style guidelines for Python. Functions were maintained at single responsibility with each function performing exactly one logical operation. Variable and function names follow snake_case convention, with descriptive names reflecting purpose rather than implementation detail. All constants such as target latency budgets and image dimensions were defined as named constants at module level, avoiding magic numbers in function bodies.

TensorFlow model compilation scripts follow Google's TFLite best practices guide for quantization-aware training and post-training quantization. TensorRT optimization scripts follow NVIDIA's Developer Best Practices for Deploying Models on Jetson. All configuration parameters were externalized to YAML configuration files to enable reproducibility without code modification. Benchmarking scripts logged timestamps, hardware utilization, and inference outputs to structured CSV files for post-hoc analysis.

5.3 Testing Standards

Testing followed IEEE Std 829-2008 (Standard for Software and System Test Documentation). Test cases were documented with unique identifiers, preconditions, input specifications, expected outputs, and pass/fail criteria before execution. Acceptance testing verified that measured TTFT values were

within the target regime bounds, that accuracy degradation from quantization was below the 2% threshold, and that power consumption did not exceed the device power envelope.

Statistical validity was ensured by reporting latency as the median of 100 inference passes rather than the mean, to reduce sensitivity to outliers caused by OS scheduling interruptions. Standard deviation and 95th-percentile latency were also recorded for all measurements. All results are reproducible using the published experimental scripts and hardware configurations documented in Section 3.3.1.

Chapter 6

Conclusion and Future Scope

6.1 Conclusion

This project has established a hardware-aware framework for selecting and optimizing vision encoders in edge-deployed Visual Large Language Models. The core contribution is the demonstration that the conventional approach of evaluating encoders solely on accuracy or FLOPs is insufficient for edge deployment; hardware-specific constraints including memory bandwidth, operator support, and thermal behavior fundamentally alter the performance landscape.

Three design principles emerge from the combined complexity analysis, Roofline modeling, and experimental validation. The Law of Resolution establishes that latency scales quadratically with image side length for CNN encoders and quartic for Vision Transformers, making input resolution a more impactful design variable than model depth for detail-sensitive tasks. The Law of Depth vs. Width establishes that edge NPU latency is dominated by memory access overhead rather than compute, making shallow-and-wide architectures systematically superior to deep-and-narrow architectures on constrained hardware. The Hybridization Principle establishes that pure ViTs are impractical for real-time edge inference with current accelerator generations, while hybrid CNN-Transformer architectures provide the Pareto-optimal trade-off between latency and accuracy.

The encoder selection matrix derived from these principles provides actionable guidance: hybrid architectures for sub-15 ms real-time requirements, EfficientNet-B0 to B3 for 15-50 ms interactive requirements, and EfficientNet-B4 for 50-200 ms batch processing on strong edge GPUs. Critically, the projector architecture must be co-optimized with the encoder; linear/MLP projectors that forward 576-1024 tokens can dominate total TTFT even with fast encoders, making Q-Former or hierarchical compressors essential for practical edge VLLMs.

6.2 Future Scope

Several promising directions for extending this work are identified. First, encoder-projector-LLM co-design represents an important frontier: rather than optimizing each component independently, joint optimization of the encoder output dimensionality, projector compression ratio, and LLM context length can yield system-level TTFT improvements not achievable through component-wise optimization. Emerging architectures such as LiteVLM and Fourier-VLM offer promising starting points.

Second, extreme quantization using INT4 and binary weight representations combined with hardware-aware mixed-precision training may enable new operating points below the 5 ms TTFT boundary for IoT platforms without sacrificing meaningful accuracy. Third, dynamic resolution adaptation as a runtime control mechanism deserves investigation as a principled approach to managing the accuracy-latency trade-off in applications with variable scene complexity. Finally, as next-generation edge NPUs are expected to include native Softmax and attention acceleration, periodic re-evaluation of the Hybridization Principle will be necessary to track the evolving hardware landscape.

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INDIVIDUAL CONTRIBUTION REPORT

Vision Encoder Optimization for Edge Visual LLMs

Architectural Principles for Hardware Selection

YASHVARDHAN SINGH

Roll No: 2206065

Abstract: This project investigates hardware-aware vision encoder selection for edge-deployed Visual Large Language Models. Three design principles governing latency behavior on resource-constrained hardware are established and validated experimentally.

Individual Contribution and Findings: Yashvardhan served as the first author and lead of the research framework design. He was responsible for conceptualizing the core research problem, defining the Sensory Bottleneck Hypothesis, and driving the end-to-end structure of the paper. As the primary architect of the theoretical foundation, he derived the formal algorithmic complexity analysis for CNN and ViT scaling behavior, formalized the Law of Resolution (including the quartic vs. quadratic complexity derivation), and framed the three design principles that constitute the central contribution of the work. He also led all experimental planning on the Raspberry Pi 5 platform, coordinated hardware procurement and lab setup, and was responsible for integrating individual contributions into a coherent research narrative. His role was analogous to a principal investigator on an academic research project.

Individual Contribution to Project Report Preparation: Authored Chapters 1 (Introduction) and 2 (Literature Review) in their entirety, including all subsections on CNN evolution, ViT complexity analysis, and hybrid architectures. Also authored Chapter 6 (Conclusion and Future Scope) and the Abstract. Responsible for the overall report narrative, ensuring consistency of argument and terminology across all chapters.

Individual Contribution for Project Presentation and Demonstration: Prepared and delivered the primary presentation slides covering system architecture, the theoretical analysis, and design principles. Led the live project demonstration for the Raspberry Pi 5 inference pipeline and presented the encoder selection matrix findings to the evaluation panel.

Full Signature of Supervisor: _____ Full Signature of Student: _____

ANKIT KUMAR

Roll No: 2206009

Abstract: This project investigates hardware-aware vision encoder selection for edge-deployed Visual Large Language Models. Three design principles governing latency behavior on resource-constrained hardware are established and validated experimentally.

Individual Contribution and Findings: Ankit served as co-author with primary ownership of the implementation and technical documentation of the paper. His core contribution was translating the theoretical principles established by the research lead into precise, accurately formatted technical content. He was responsible for implementing the LaTeX formatting of the paper draft, typesetting all mathematical equations (including the MHSA complexity derivation, Roofline model equations, and TTFT decomposition formula), and constructing all tables and figures to publication standard. He conducted the NVIDIA Jetson Orin Nano experimental benchmarks, compiled models using TensorRT, and was responsible for the accuracy of all numerical results appearing in the paper. He also drafted the Implementation chapter (Chapter 4) in collaboration with the team.

Individual Contribution to Project Report Preparation: Authored Sections 4.1, 4.2, and 4.3 of the Implementation chapter. Prepared and formatted all tables (Tables I through IV) and figure

captions. Responsible for equation formatting throughout the document and for ensuring numerical consistency between in-text claims and tabulated results.

Individual Contribution for Project Presentation and Demonstration: Designed the technical slides for the experimental results section. Delivered the demonstration of the Jetson Orin Nano inference pipeline and walked the panel through the latency-accuracy trade-off analysis.

Full Signature of Supervisor: _____ Full Signature of Student: _____

ANANYA SINGH

Roll No: 2206008

Abstract: This project investigates hardware-aware vision encoder selection for edge-deployed Visual Large Language Models. Three design principles governing latency behavior on resource-constrained hardware are established and validated experimentally.

Individual Contribution and Findings: Ananya led the literature review process for the entire project, which formed a critical foundation for validating the novelty and positioning of the research. She was responsible for identifying, reading, and synthesizing all 37 references cited in the paper, ensuring that claims made in the theoretical framework were grounded in existing published work. She performed systematic fact-checking across the manuscript, verifying that benchmark numbers, hardware specifications, and architectural descriptions were accurate with respect to source papers. She also cross-referenced the PolyThrottle findings against the paper's claims and validated the Hybridization Principle by locating and reviewing the original EfficientFormer and FastViT benchmarks. In addition, she led the requirements analysis section and the system architecture description in Chapter 3.

Individual Contribution to Project Report Preparation: Authored Chapter 3 (Problem Statement / Requirement Specifications) in full, including the project planning, requirements analysis, design constraints, and system pipeline architecture sections. Authored Section 2.1 (Evolution of Vision Encoders) and contributed to Section 2.3 (PolyThrottle Framework). Responsible for reference validation and bibliography formatting.

Individual Contribution for Project Presentation and Demonstration: Presented the literature review and related works section to the evaluation panel. Demonstrated the Coral TPU inference pipeline and explained the hardware constraints driving architectural choices.

Full Signature of Supervisor: _____ Full Signature of Student: _____

SHUBHANGI KUMARI

Roll No: 2206128

Abstract: This project investigates hardware-aware vision encoder selection for edge-deployed Visual Large Language Models. Three design principles governing latency behavior on resource-constrained hardware are established and validated experimentally.

Individual Contribution and Findings: Shubhangi served as the reviewer and quality gatekeeper across all stages of the project and paper. Her role was to verify the correctness of every section at each draft milestone, identifying logical inconsistencies, unsupported claims, ambiguous terminology, and formatting deviations before the document advanced to the next stage. She reviewed the theoretical derivations for mathematical accuracy, cross-checked the experimental results against raw measurement logs, and verified that all figures and tables were internally consistent with the text. She also took ownership of the Standards Adopted chapter, ensuring that all IEEE and industry standards cited were applicable and accurately described. Her review function was essential to the paper reaching publication-ready quality.

Individual Contribution to Project Report Preparation: Authored Chapter 5 (Standards Adopted) in full. Performed multi-pass review of all other chapters, providing written correction notes at each

iteration. Responsible for final proofreading of the complete report prior to submission, including consistency of formatting, numbering, cross-references, and compliance with the KIIT report format guidelines.

Individual Contribution for Project Presentation and Demonstration: Coordinated the overall presentation structure and slide design. Delivered the quality assurance and standards section of the presentation. Responsible for reviewing and approving the final slide deck before submission.

Full Signature of Supervisor: _____ Full Signature of Student: _____

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